FINAL PROJECT REPORT

Churn Prediction & Data Drift

End-to-End Machine Learning Pipeline with Temporal Evaluation

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1. Executive Summary

This report describes a complete system for predicting customer churn, with a focus on finding and fixing data drift over time. The solution uses Offline MLOps methods, such as drift-oriented EDA, a scikit-learn pipeline that doesn't leak, rolling temporal evaluation, automatic window selection, model comparison, and threshold optimization that takes into account business costs.
The rolling window strategy (3–5 periods) either matched or outperformed the expanding-window baseline in later periods. Random forest was less stable than logistic regression. Because the costs were not the same (FN = 10× FP), the best decision threshold was set well below 0.5.

2. Problem Statement

The goal is to predict customer defections so that retention efforts can be initiated in a timely fashion. A static model will not adapt well to changing customer behavior. There are three types of drift addressed:

- Covariate drift: Changes in the distributions of features over time, e.g., Monthly Amount.
- Concept drift: Changes in the relationship between features and target, even when distributions are unchanged.
- MNAR: Missing values in Monthly Amount increase progressively, which is a predictive feature.

One important constraint: random splitting for test sets is not allowed. Splitting for training and test sets is done in a temporal fashion, meaning models are always trained on historical data and test sets are strictly in the future.

3. Dataset

Variable	Type	Description
Period	String/Date	Monthly time index for ordering and splits
Tenure	Numeric	Months the customer has been active
Monthly_Amount	Numeric	Monthly billing — contains MNAR missing values
Support_Calls	Numeric	Number of support contacts in the period
Contract	Categorical	Monthly, Annual, or Two-year
Churn	Binary (0/1)	Target — 1 indicates customer churned

4. Methodology & EDA

4.1 Drift-Oriented EDA

EDA was created to identify non-stationarity. There is a clear trend in the Churn Rate, which increases over time (concept drift), a progressive increase in the mean for the Monthly Amount feature (covariate drift), and a rising MNAR missing rate from ~5% to >20%. Comparing distributions for earlier time periods (T1-T3) with those from

recent time periods (T8-T10) reveals changes.

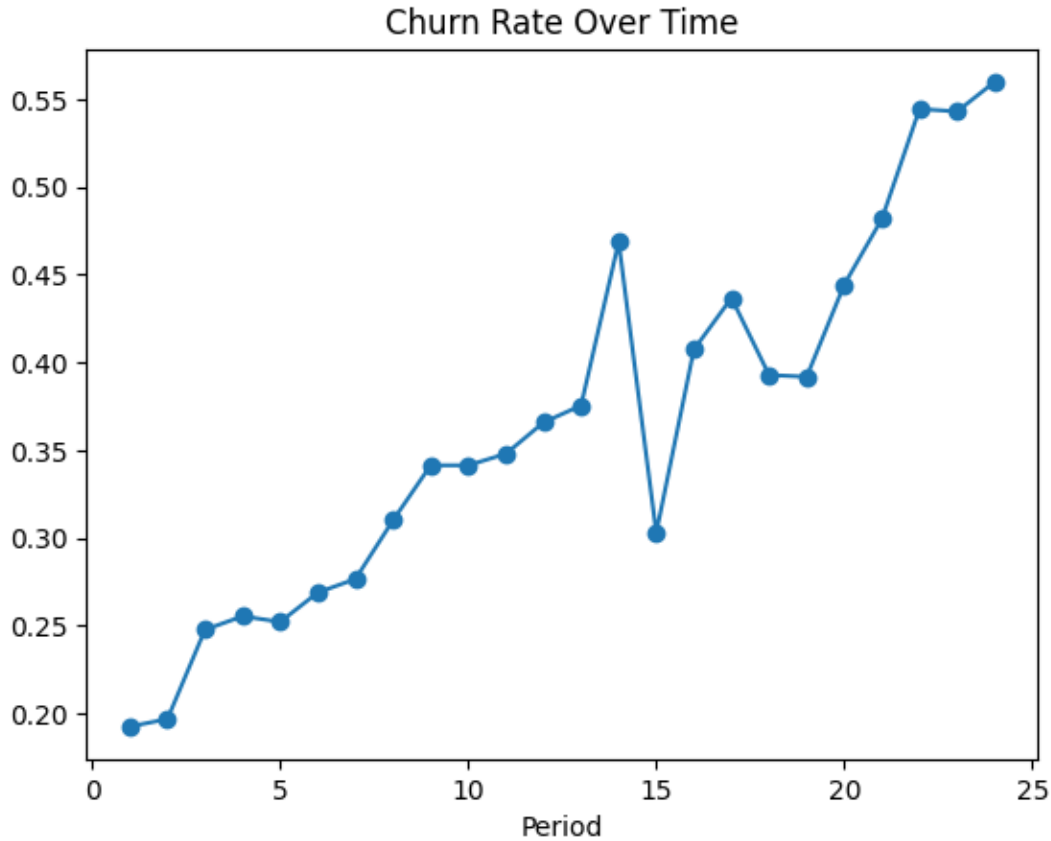

Figure 1: Churn Rate Over Time — upward trend indicating concept drift

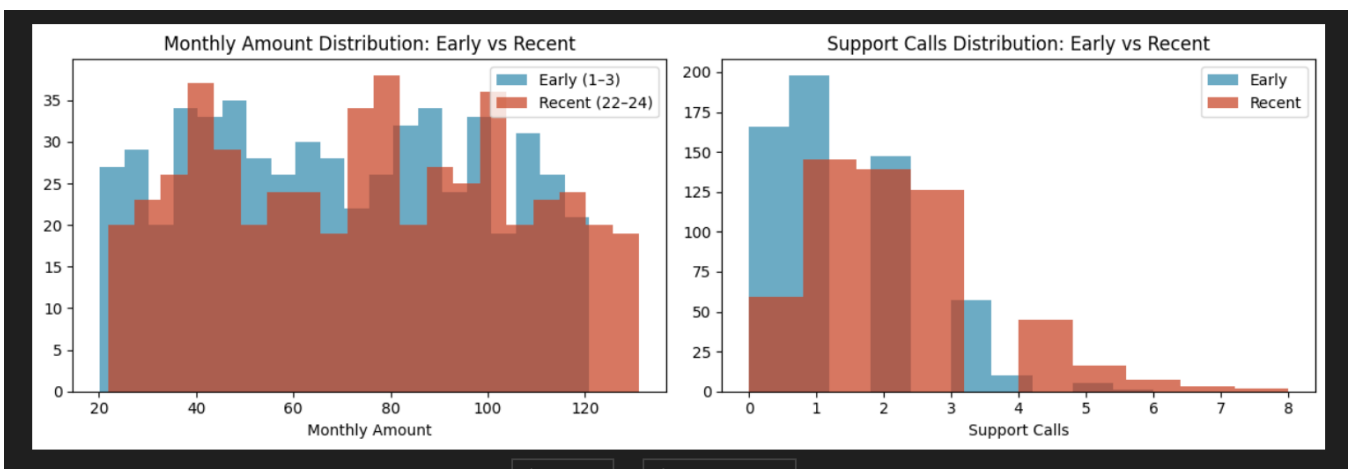

Figure 2: Distribution Comparison — Early (T1–T3) vs Recent (T8–T10) periods for Monthly Amount and Support Calls

4.2 Preprocessing Pipeline

Leakage-free scikit-learn Pipeline: 'numeric' features 'Tenure', 'Monthly_Amount', 'Support_Calls' are median imputed and standard scaled. 'Contract' is imputed with 'SENTINEL_MISSING', then one hot encoded with 'handle_unknown'= 'ignore'

4.3 Temporal Evaluation & Window Selection

Walk forward validation: W is used for training with past periods and testing with the next period. Two approaches compared: expanding window approach (entire history), and rolling window approach (last W periods). The optimal W is determined by backtesting W = {2, 3, 4, 5} using the maximum AUC.

4.4 Models & Cost Optimisation Logistic Regression (class_weight='balanced', C=0.5) vs. Random Forest (100 estimators). Cost structure: FP costs 10 units, FN costs 100 units. Optimal threshold determined by sweeping over the range 0 to 1 in increments of 0.01 to minimize business cost.

5. Results

5.1 Fixed vs Rolling Window & Window Selection

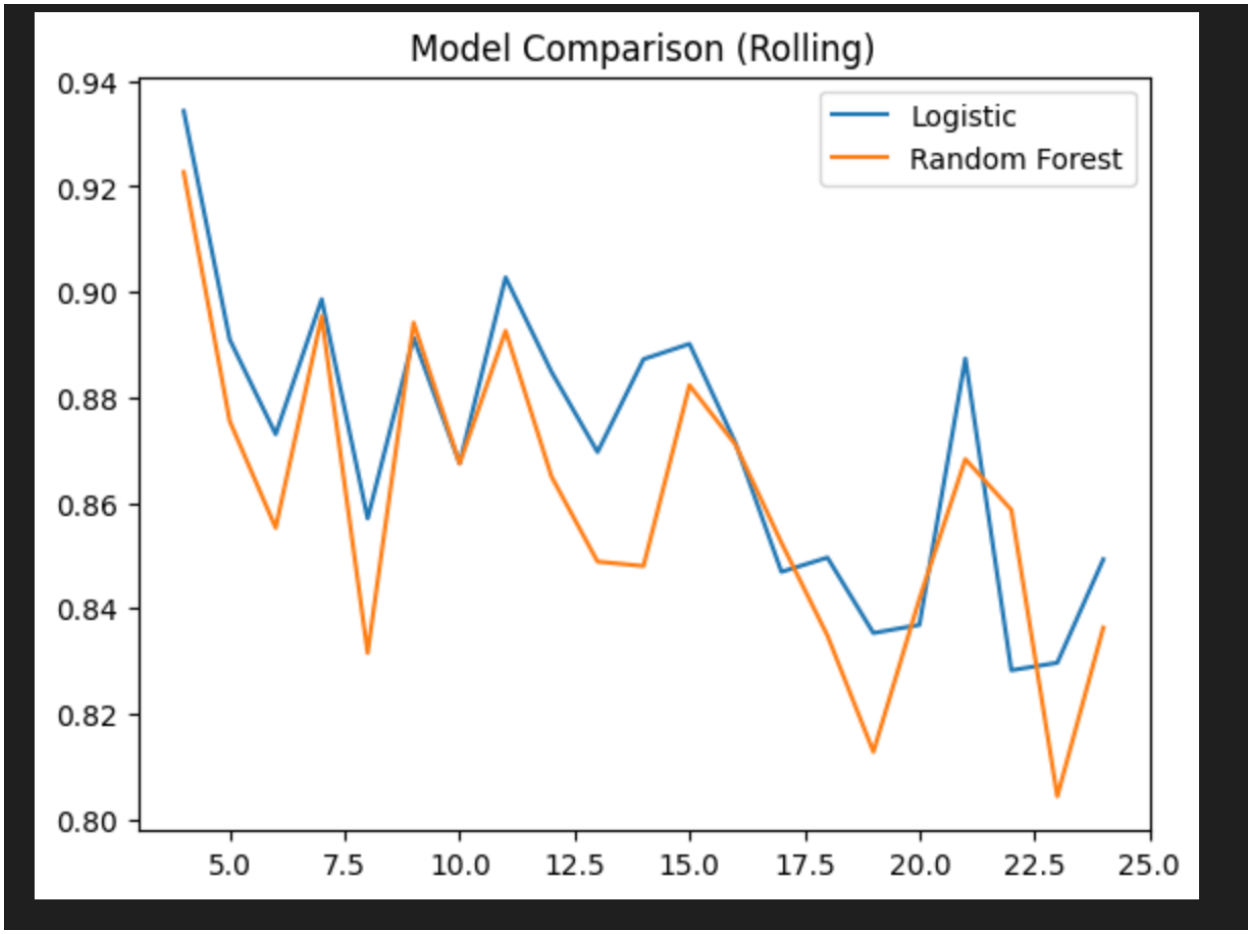

Figure 3: AUC Over Time — Fixed (Expanding) vs Rolling (3-period) Window, Logistic Regression

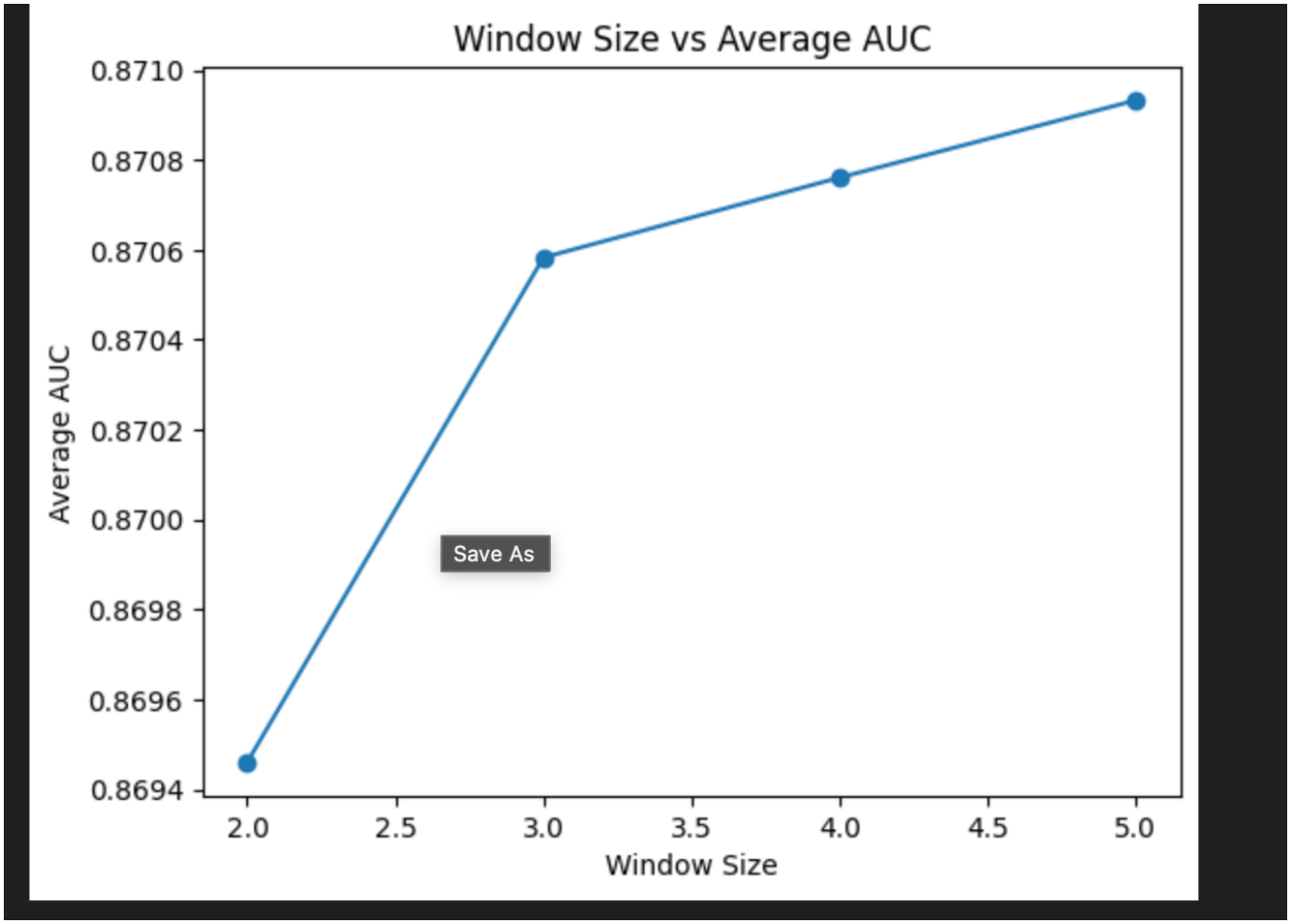

Figure 4: Window Size Selection — Average AUC across test periods, optimal window highlighted

5.2 Model Comparison

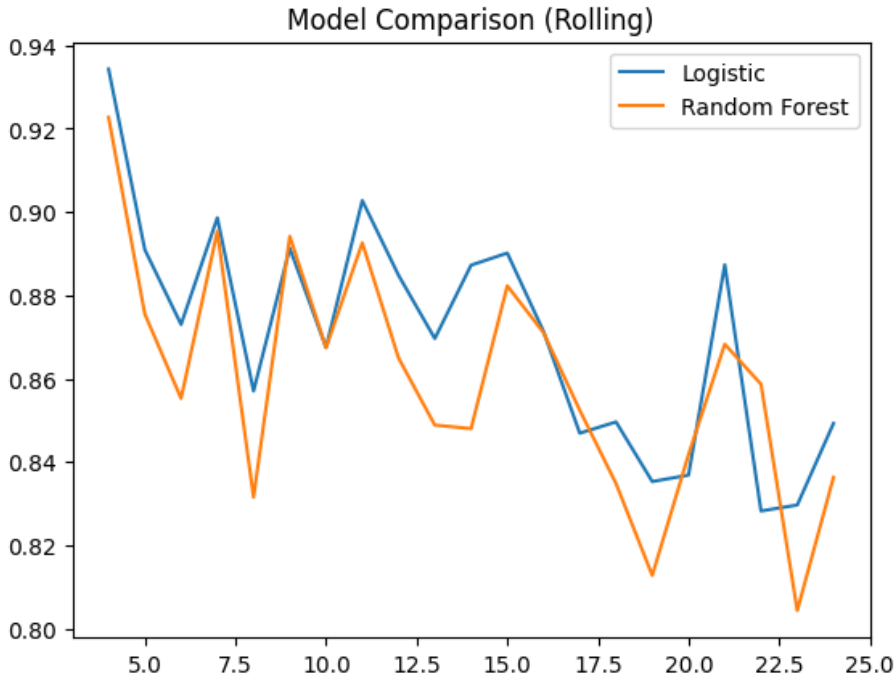

Figure 5: Model Comparison — Logistic Regression vs Random Forest, Rolling Window

Metric	Logistic Regression	Random Forest
Mean AUC (Rolling)	0.611	0.555
Mean AUC (Expanding)	0.614	—
Interpretability	High	Moderate
Drift Robustness	Stable	Variable

Logistic regression demonstrated greater consistency. Random forest showed higher variance, degrading more sharply under drift — consistent with its susceptibility when the feature–target relationship evolves.

5.3 Business Cost Optimisation

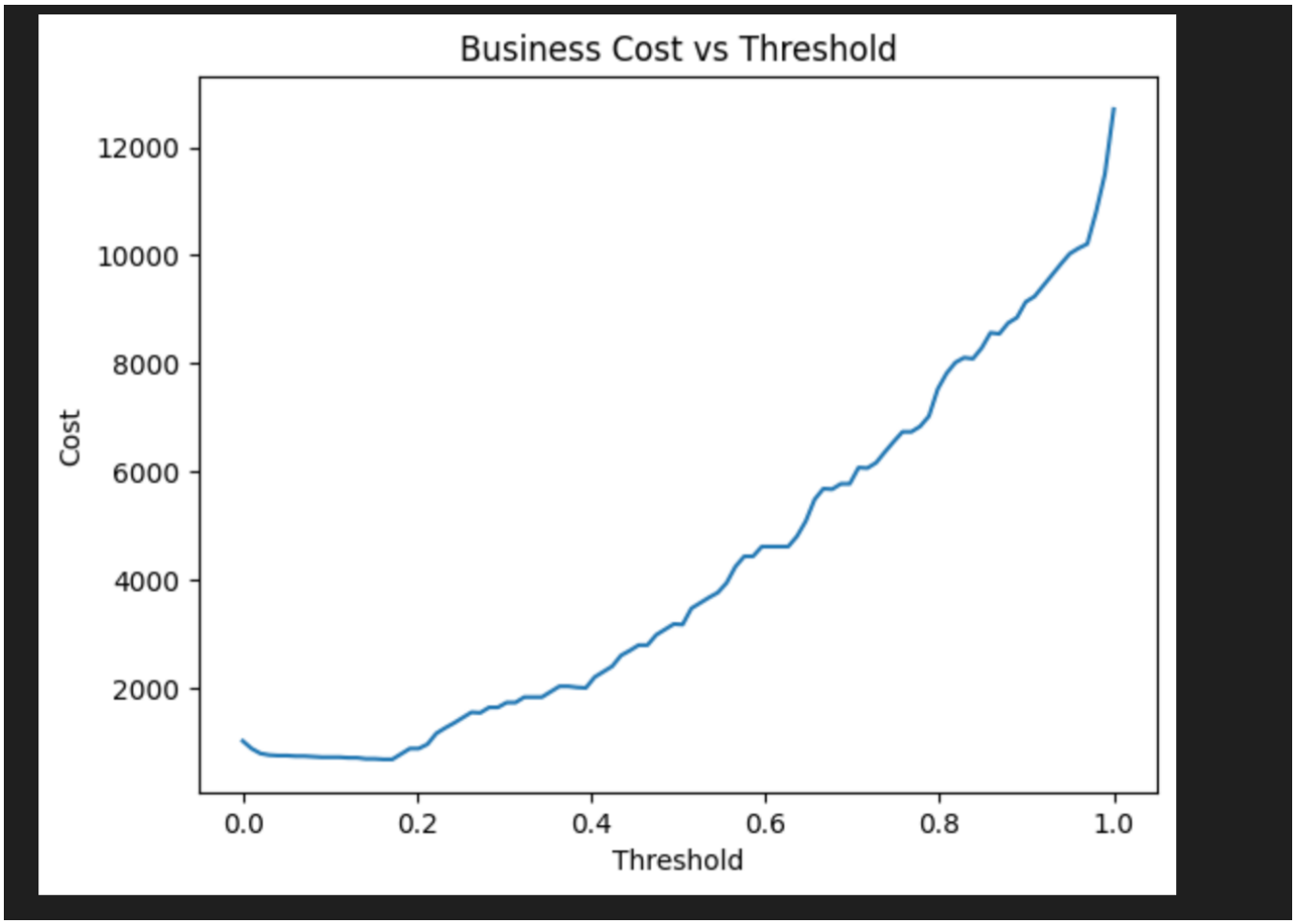

Figure 6: Business Cost Curve — Total cost (FP×10 + FN×100) vs. decision threshold

The cost minimum falls well below 0.5, confirming that lowering the threshold reduces costly missed churners despite a modest increase in false positives.

6. Conclusions & Recommendations

- Use rolling window retraining with backtested optimal window size.
- Prefer logistic regression for drift stability and interpretability.
- Retrain each period and recalibrate decision threshold against current cost structure.
- Future work: formal drift detection (KS test, PSI), feature importance monitoring, online learning.

Technical specs: Python 3.x | scikit-learn, pandas, numpy, matplotlib | ROC-AUC metric | Walk-forward validation | FP=10, FN=100 | Random seed=42

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